Matthew D. Eichner

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Education

University of California, Berkeley *Aug. 2026 – Present*
Ph.D. in Statistics
- Diversity and Community Fellow

Pomona College *May 2026*
Bachelor of Arts in Mathematics, Cum Laude GPA: 3.96/4.0
- Bruce Jay Levy Senior Prize in Mathematics
- Distinction in the Senior Exercise
- Nominee, Sigma Xi – The Scientific Research Honor Society (2026)
- Pomona Scholar Designation (FA 2022, FA 2023, SP 2024, FA 2024, SP 2025)

Research Experience

Senior Thesis — Number Theory & Dynamical Systems *Claremont, CA*
Pomona College *Jan. 2025 – May 2026*
Advisor: Dr. Taylor McAdam, Pomona College
- Define and prove the existence of a novel infinite integer sequence derived from the reduced fraction-valued endpoints of the Cantor set, linking classical fractal constructions to discrete number-theoretic structures.
- Extend analysis to secondary sequences generated from the endpoints, employing combinatorial, number-theoretic, and computational tools to bridge the study of fractal geometry with the theory of integer sequences.
- Generalize results to sequences arising from alternative Cantor set constructions (e.g., middle-half, fat, Smith–Volterra, and stochastic variants), demonstrating the robustness and universality of the derived patterns across families of similarly derived fractals.
- Publish resulting sequences to the Online Encyclopedia of Integer Sequences (A390119) and preparing a forthcoming paper detailing theoretical results and generalizations.

Summer Undergraduate Researcher — Population Dynamics *Chicago, IL*
NSF-Simons National Institute for Theory and Mathematics in Biology *June 2025 – Aug. 2025*
Advisors: Dr. Alvin Bayliss, Dr. Vladimir Volpert, Northwestern University
- Modeled and explored networks of rock–paper–scissors systems to investigate how coupling strength influences stability, coexistence, and pattern formation in competitive ecosystems.
- Simulated inter-species dynamics using nonlinear differential equations and custom Python-based Improved Euler methods, developing computational tools for three species systems.
- Leveraged high-performance computing to generate and analyze large-scale time series, identifying transitions between synchronized, desynchronized, and intermittent states that reveal critical thresholds in system behavior.
- Applied spectral analysis and Fourier transforms to characterize quasi-periodic and chaotic regimes, contributing to a deeper understanding of deterministic chaos in ecological and mathematical models.
- Presented research findings at the inaugural NITMB MathBio Convergence Conference, engaging with the broader mathematical biology community on the intersection of nonlinear dynamics and biological systems.

RampDB **Lead Developer** *New York, NY*
Columbia University Mailman School of Public Health *July 2024 – May 2025*
Advisor: Dr. Wenpin Hou, Columbia University

- Initiated the development of an interactive and publicly available Predicted DNA Methylation (DNAm) database to facilitate research in epigenomic mechanisms and therapy.
- Utilized dimension reduction techniques (i.e. PCA, UMAP) with high power computation and visualization software to analyze bulk, single-cell, and spatial RNA-seq data in order to reconstruct the DNA methylation landscape.
- Designed and implemented user-friendly online interface to host DNAm data in the form of an R Shiny App, facilitating deeper insights into gene regulation for researchers.

Participant — Summer Institute in Biostatistics & Data Science *New York, NY*
Columbia University Mailman School of Public Health *May 2024 – July 2024*
Advisor: Dr. Wenpin Hou, Columbia University

- Developed and applied quantitative and computational skills in biostatistics and data science, gaining comprehensive training in R for data analysis, visualization, and reproducible research.
- Attended seminars on advanced biostatistical methods, research design, and ethical conduct, deepening understanding of real-world challenges in biomedical data analysis.
- Conducted a mentored NIH-funded data analysis project employing dimension reduction and clustering algorithms to uncover structure in high-dimensional datasets, culminating in a formal research presentation at the program colloquium.

Leadership Experience

Mathematics Department Mentor *Claremont, CA*
Pomona College Dept. of Mathematics *August 2023 – Present*

- Courses: Statistical Theory, Probability, Linear Algebra, Introduction to Statistics, Calculus II.
- Clarify and explain topics from class and homework through self-developed supplemental lecture and practice exercises for full class sections of 20+ students.
- Organize weekly review sessions and team building activities within the classroom environment.
- Guide students with developing successful study strategies for mathematics and statistics courses.
- Assist professor with curriculum development.

CARES Campus Advocate *Claremont, CA*
Pomona College Title IX and CARES Office *August 2023 – Present*

- Organize and facilitate Title IX education events through collaboration with other student and administrative bodies to cultivate a culture of inclusion and survivor-centered dialogue.
- Develop and present curricula for gender-based violence prevention and bystander awareness to provide comprehensive first-year student training.
- Serve as a confidential resource providing direct support to students affected by sexual violence and related harm with extensive training through the CARES Office and community partners in order to provide trauma-informed support.

Activities

Pomona-Pitzer Varsity Baseball Team *Aug. 2022 – Dec. 2024*
Pomona-Pitzer Baseball Broadcast Commentator *Feb. 2025 – May 2025*
Language: Spanish—Conversational
Certifications: Responsible Conduct of Research—Biomedical